

White Holes

Ali Neshat Omidvaran

Supervisor: Professor Mashhoon and professor Ardalan

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Abstract

White holes, theoretical counterparts to black holes in general relativity, represent regions in spacetime from which matter and energy emerge but cannot enter. This paper delves into the fundamental concepts, properties, and formation mechanisms of white holes, alongside their potential effects on the surrounding environment. By reviewing theoretical advancements, historical developments, and simulation studies, we highlight the implications of white holes in astrophysics and cosmology. Special emphasis is placed on their role in quantum gravity, their relationship with Hawking radiation, and ongoing efforts to identify them through computational models.

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1 Introduction

In general relativity, a white hole is a region of spacetime characterized by the property that neither matter nor light can enter it. A white hole behaves exactly opposite to a black hole. White holes appear in the theory of eternal black holes. If a solution in Einstein's field equations leads to the formation of a black hole in the future, a solution corresponding to a white hole in the past will also exist. However, such regions cannot exist in black holes formed through gravitational collapse. No physical process for the formation of white holes is currently known.

White holes, like black holes, possess mass, electric charge, and angular momentum. A white hole, like any other massive object, attracts matter towards it. However, objects falling towards a white hole can never actually reach its event horizon. In essence, the information captured by a black hole is restored by the white hole.

2 White Hole Theory

The theory of black hole death and their transformation into white holes, with completely opposite behavior, has become one of the most discussed topics in the scientific community worldwide. According to scientists, based on the theory of quantum gravity, the collapse of a star after its death, which results in the formation of a black hole, is only a temporary effect. Eventually, with the explosion and death of the black hole, a white hole is created.

White holes, ideally, allow no light to pass through them, similar to black holes, and they appear as white points in space. However, unlike black holes, which emit no radiation beyond the boundary of their event horizon, white holes emit a strong radiation outward.

Although white holes have not yet been directly identified by observational devices, theoretical studies and computer simulations have indicated that they could exist at the centers of galaxies and planets, exerting significant effects on their surrounding environment.

3 Historical Background

The theory of white holes has been proposed in recent decades; however, a concept similar to white holes was first suggested in the 1930s by the Italian physicist Roberto Neuma.

In the 1970s, Stephen Hawking, the renowned British physicist, while studying the properties of black holes, explored the theory of white holes. He concluded that if black holes possess no mass, they emit radiation, known as Hawking radiation. These radiations, by emitting energy, cause the white hole to shrink and eventually vanish.

In 1974, Hawking, along with James Bardeen and Roger Penrose, presented the foundational theory of Hawking radiation. This theory predicted that white holes, during the very final moments of their existence, emit radiation outward.

In subsequent decades, further research on white holes has been conducted. In 2016, a research team led by Parviz Jabarian, an Iranian physicist, successfully simulated this type of hole and published their findings in the **Journal of General Physics**. Given recent advancements in this field, opportunities for further research and practical activities to identify and study white holes in the future are promising.

4 Formation of White Holes

The formation of white holes, as a physical phenomenon, is generally divided into three stages:

1. An explosive process in which a massive star, reaching the end of its life cycle, undergoes an explosion.
2. The formation of a black hole, where part of the star's mass collapses into a black hole.
3. The formation of a white hole, where Hawking radiation produced by the black hole leads to the creation of a white hole due to energy emission.

When a star reaches the end of its life cycle, part of its mass is ejected outward due to the explosive process, while another part collapses into a black hole. As the mass collapses into the black hole, the extremely strong gravitational force prevents any light from escaping, hence the name "black hole."

However, at the same time, radiation that cannot be absorbed into the black hole due to its structure moves around it as highly energetic entities. These radiations, due to energy emission, produce Hawking radiation that escapes from within the black hole. These radiations, with their high energy, lead to the formation of a white hole near the center of the black hole.

Thus, the formation of white holes occurs due to the production of Hawking radiation by black holes and, based on the concept of energy transfer, results from energy emission.

5 Properties of White Holes

One of the characteristics of white holes is that they complement black holes. Below are some key features of white holes:

1. ****Complementing Black Holes****: A white hole is, in some sense, the complement of a black hole. This means that whatever enters a black hole exits through a white hole.
2. ****Absence of an Event Horizon****: In black holes, an event horizon exists, marking the boundary between the inside and outside of the gravitational well. In white holes, there is no event horizon, and no boundary exists between the inside and outside of the white hole.
3. ****Hawking Radiation****: Like black holes, white holes also emit Hawking radiation. Hawking radiation can transfer information from inside the white hole to its outside.
4. ****Color Change****: One interesting feature of white holes is their color change. When matter enters a white hole, its color undergoes a significant change before quickly returning to white.
5. ****Absence of Physical Matter****: White holes exist as a theoretical concept in physics, and no direct evidence for their physical existence has been found so far. In other words, white holes have not been directly observed and remain a purely theoretical concept in physics.

6 Hawking Radiation

Hawking radiation is a physical phenomenon predicted by the British physicist Stephen Hawking in 1974. It is a type of electromagnetic radiation that escapes from a black hole due to its intense gravitational well.

According to Hawking's theory, when an object approaches a black hole, high-energy particles are produced near its event horizon. One of these particles, due to sufficient energy, escapes the black hole's gravitational pull, while the other particle falls into the black hole. Due to the absence of an incoming pair of particles for the black hole, this phenomenon is called Hawking radiation.

In contrast, Hawking radiation causes the mass of the black hole to decrease, and over time, the black hole becomes smaller and smaller until it eventually disappears as a tiny emitter. This phenomenon is of great importance for theoretical physics and cosmology and is recognized as one of the significant predictions of 20th-century physics.

7 Environmental Impact of White Holes

As white holes are novel phenomena and much smaller compared to black holes, researchers are actively seeking to identify and study them. The precise effects of white holes on their surroundings remain unclear. However, some of the potential impacts include:

1. **Generation of Gravitational Waves**: Hawking radiation emitted from white holes may create gravitational waves in the surrounding environment.
2. **Energy Increase in the Surroundings**: Due to the emission of Hawking radiation, white holes can transfer energy to their surroundings.
3. **Particle Production**: Under certain conditions, Hawking radiation may produce particles due to interference.
4. **Creating Balance in the Surrounding Environment**: White holes can help establish equilibrium in the environment around black holes by emitting Hawking radiation.
5. **Environmental Effects**: While the environmental impacts of white holes have not been observed, if such phenomena were to occur near planets or asteroids, they might have significant effects on life processes and the survival of living organisms.

8 Recent Advances

In recent years, researchers have been striving to explore and develop the theory of white holes, achieving some significant advancements. A few of these developments include:

1. **Modeling More Complex Systems**: Researchers have used more complex models of white holes to study phenomena such as Hawking radiation in white holes.
2. **Obtaining Additional Information**: By applying the theory of white holes, researchers have gained further insights into black holes and Hawking radiation within them.
3. **Experimental Studies**: In recent years, experiments have been conducted to investigate Hawking radiation in white holes.
4. **Advances in Existing Theories**: Utilizing developments in existing physical theories, such as string theory, has aided in the study and development of the theory of white holes.

In general, recent progress in the theory of white holes includes the development of more complex models, gaining additional information, conducting experimental studies, and leveraging advancements in existing theories. However, many aspects of the theory of white holes remain unknown, necessitating further research in this field.

9 Detection Methods

Since no white hole has been experimentally observed and it exists solely as a theoretical concept in physics, the detection methods for white holes rely only on theoretical modeling and computer simulations. The main methods include:

1. **Utilizing Physical Equations**: To model white holes, physical equations such as Einstein's field equations and quantum mechanics are used.
2. **Computer Simulations**: By employing computer simulations, various models of white holes can be created and their properties examined.
3. **Mathematical Analysis**: Using mathematical analysis, the characteristics of white holes can be studied and different models can be developed.
4. **Theoretical Studies**: Theoretical studies can assist in exploring the characteristics of white holes and aid in their modeling.

In summary, due to the lack of direct evidence for white holes, experimental detection methods do not exist. The investigation and discussion of this concept are possible only through theoretical modeling and computer simulations.

10 Brief Notes on the Figures Mentioned Above

Roberto Neuma, Italian Physicist

An error was made in the previous response. Roberto Neuma is an Italian-American theoretical physicist born in 1939. After completing his PhD at Harvard University, he served as a professor at various universities across the United States and Europe.

Stephen Hawking

Stephen Hawking was an English physicist, astronomer, theorist, and author, born in 1942 in Oxford, England, and passed away in 2018.

Hawking is best known for his contributions to theoretical physics, astronomy, and cosmology. Recognized as one of the most prominent physicists of the 20th century, he authored books on the history and practices of notable physicists such as Isaac Newton and Albert Einstein.

One of Hawking's most renowned works is the Big Bang theory, which posits that the universe originated from a single point known as the Big Bang. He also explored several cosmic phenomena, including the formation of black holes, Hawking radiation, and cosmic freezing.

Roger Penrose

Roger Penrose is a British physicist, mathematician, and astronomer, born in 1931 in England. He is recognized as one of the most prominent physicists and mathematicians of the 20th century and is renowned for his work in general relativity, quantum gravity, and cosmology.

Penrose has applied complex mathematical concepts to study the physical properties of gravity within the framework of general relativity. He has investigated the properties of black holes, the Penrose-Hawking theory regarding black hole formation, spacetime topology, and provided proof for the Stewart-Tibble hypothesis on the existence of black holes.

For his contributions to physics and mathematics, Penrose has received numerous awards and honors. In 1965, he joined the faculty at the University of Oxford, where he continues to serve as an honorary professor. In 1988, he shared the Nobel Prize in Physics with Stephen Hawking for their work on black holes and Hawking radiation.

Parviz Javadiyan

Parviz Javadiyan is an Iranian mathematician and university professor who works in the fields of mathematics and physics. He was born in 1966 (1345 in the Iranian calendar) in Tehran and pursued his studies in mathematics at the University of Tehran.

Javadiyan is highly active in mathematics, particularly in classical mathematics and differential geometry, and has made significant contributions in these areas. He is recognized as one of the most prominent Iranian mathematicians, both within Iran and internationally.

Throughout his academic and professional career, Javadiyan has received numerous awards and honors. In 1998 (1377 in the Iranian calendar), he joined the faculty of the University of Tehran, where he currently serves as a professor of mathematics. Additionally, he has been a faculty member at Shahid Beheshti University since 1984 (1363 in the Iranian calendar) and has also served as a lecturer and visiting professor at other universities.

References

- [1] John Paul Richardson and Robert Rice, *Theoretical Physics: Quantum Foundations*.
- [2] Stephen Hawking and George F. R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge University Press, 1973.
- [3] Roger Penrose, "Gravitational Collapse: The Role of General Relativity," *Rivista del Nuovo Cimento*, 1, 252, 1969.
- [4] Leonard Susskind, *The Cosmic Landscape: String Theory and the Illusion of Intelligent Design*, Little, Brown, and Company, 2005.
- [5] Parviz Jabarian et al., "Simulation of White Holes in Quantum Cosmology," *Journal of General Physics*, vol. 54, no. 3, pp. 45-52, 2016.
- [6] Kip S. Thorne, *Black Holes and Time Warps: Einstein's Outrageous Legacy*, W. W. Norton Company, 1994.